MIL-B-29165A(YD)
22 October 1982
SUPERSEDING
MIL-B-29165(YD)
2 June 1976

MILITARY SPECIFICATION

BUILDINGS, PREFABRICATED, PANELIZED: WOOD, ONE-ROOM, ARCTIC CLIMATE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers prefabricated, frameless buildings having straight side and end walls, a gable roof, and castella type steel foundation beams. The buildings are designed for use in arctic climates. Buildings are to be rapidly erected, dismantled, and reassembled to permit complete reuse of all prefabricated parts and assemblies.

1.2 Classification. Prefabricated wood buildings shall be of the following types and styles, as specified (6.2.1).

Type I - Personnel building, 28 by 128 feet.

- Style A - Personnel quarters, recreation, or administration facilities with basic foundation.
- Style B - Dispensary or galley with special foundation.
- Style C - Utility building with special foundation.

Type II - Maintenance building, 28 by 56 feet.

- Style A - Equipment maintenance building.
- Style B - Storage building.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5410

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

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SPECIFICATIONS

FEDERAL

MM-L-736 - Lumber; Hardwood.
NN-P-530 - Plywood, Flat Panel.
QQ-S-781 - Strapping, Steel, and Seals.
UU-B-1780 - Building Paper, Vegetable Fiber or Inorganic Fiber Reinforcement, Reflective.
MMM-A-187 - Adhesive, Epoxy Resin Base, Low and Intermediate Strength, General Purpose.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-C-104 - Crate, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-T-704 - Treatment and Painting of Material.
MIL-R-900 - Rubber Gasket Material, 45 Durometer Hardness.
MIL-W-3448 - Wallboard; Packaging of.
MIL-C-3774 - Crates, Wood: Open 12,000 and 16,000 Pound Capacity.
MIL-R-6130 - Rubber, Cellular, Chemically Blown.
MIL-R-14328 - Rubber Sheet: Synthetic, Medium Soft, General-Purpose Gasket Material (For Extreme Climatic Conditions).
MIL-L-14362 - Lumber, Unitizing and Loading of.
MIL-B-26195 - Boxes, Wood-Cleated, Skidded, Load-Bearing Base.
MIL-C-52950 - Crates, Wood, Open and Closed.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing: With Appropriate Test Methods.
MS35829 - Hinge, Butt, Continuous (Piano) 0.075 Thickness, 0.1878 and 0.250 Pin Diameter.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF COMMERCE (DOC)

PS20 - American Softwood Lumber Standard.

(Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

Drawing No.	Description
	Arctic Building, Wood-Prefabricated-Panelized, 28 by 128 by 10 feet.
1110482	Schedules.
1110483	Assembly Diagram.
1110484	Transverse Section.
1110485	Floor Panels.
1110486	Floor Panels.
1110487	Floor Panels.
1110488	Quarters-Utilities-Galley, Floor Panel.
1110489	Wall Panels.
1110490	Wall Panels.
1110491	Wall Panels.
1110492	Wall Panels.
1110493	Quarters-Utilities-Galley, Door Panel W-19.
1110494	Door for Panel W-19.
1110495	Wall Panel Details.
1110496	Gable Panels.
1110497	Truss Assembly Details.
1110498	Outer Roof Truss.
1110499	Center Roof Truss.
1110500	Roof Panels.
1110501	Roof Panels.
1110502	Roof Details.
1110503	Key Plan and Liner Details.
1110504	Footings.
1110505	Foundation.

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Arctic Building, Wood-Prefabricated-Panelized, 28 by 56 by 12 feet, Maintenance and storage.

1110506	Schedules.
1110507	Assembly Diagram.
1110508	Exterior Views.
1110509	Transverse Section.
1110510	Longitudinal Section.
1110511	Floor Panel.
1110512	Floor Panels.
1110513	Floor Panels.
1110514	Floor Panels.
1110515	Floor Panel Details.
1110516	Basic Floor Plan.
1110517	Wall Panel.
1110518	Wall Panel.
1110519	Wall Panels.
1110520	Wall Panels and Gable Panels.
1110521	Door Panel.
1110522	Personnel Door Details.
1110523	Wall Panels.
1110524	Wall Panel Details.
1110525	Overhead Door Installation.
1110526	Door Jamb and Header Details.
1110527	Truss and Door Assemblies.
1110528	Outer Roof Truss.
1110529	Center Roof Truss.
1110530	Truss Details.
1110531	Roof Panels.
1110532	Roof Panels.
1110533	Roof Panel Details.
1110534	Roof and Smokestack Details.
1110535	Footings, Structural.
1110536	Foundation.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC., (ANSI)

A199.01 - Construction and Industrial Plywood.
A135.4 - Basic Hardboard.
A201.1 - Screens.

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B18.1.1 - Small Solid Rivets.
B18.1.2 - Large Rivets.
B18.2.1 - Square and Hex Bolts and Screws.
B18.2.2 - Square and Hex Nuts.
B18.6.1 - Head Wood Screws.
B18.6.4 - Slotted and Recessed Head Tapping Screws and Metallic Drive Screws.
B18.22 - Plain Washers.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Specification for Structural Steel.
A242 - Specification for High Strength, Low Alloy Structural Steel.
A386 - Specification for Zinc Coating (Hot Dip) on Assembled Steel Products.
A466 - Specifications for Weldless Carbon Steel Chain.
A475 - Specification for Zinc-Coated Steel Wire Strand.
A526 - Specification for Sheet Steel Zinc-Coated (Galvanized) by The Hot Dip Process, Commercial Quality.
B209 - Specification for Aluminum-Alloy Sheet and Plate.
B308 - Specification for Aluminum-Alloy Standard Structural Shapes, Rolled or Extruded.
C591 - Specification for Rigid Preformed Cellular Urethane Thermal Insulation.
C612 - Specification for Mineral Fiber Block and Board Thermal Insulation.
D2559 - Specification for Adhesive for Structural Laminated Wood Products for Use Under Exterior (Wet Use) Exposure Conditions.
E84 - Test Method for Surface Burning Characteristics of Building Materials.
F547 - Nails for Use with Wood and Wood-Base Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WOOD PRESERVERS' ASSOCIATION (AWPA)

C20 - Structural Lumber, Fire-Retardant Treatment by Pressure Process.
C27 - Plywood, Fire-Retardant Treatment by Pressure Process.

(Application for copies should be addressed to American Wood Preservers' Association, 1625 Eye Street, N.W., Washington, DC 20006.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC., (SAE)

SAE - Handbook.

(Application for copies should be addressed to Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

WESTERN WOOD PRODUCTS ASSOCIATION (WWPA)

Standard Grading Rules.

(Application for copies should be addressed to Western Wood Products Association, Yeon Building, Portland, OR 97204.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The building shall be prefabricated and furnished unassembled, and shall include prefabricated: floor, wall, and roof panels; personnel and equipment doors; hermetically sealed windows; roof truss, assemblies, and connectors; steel beam foundation; and smoke pipe assemblies. The building shall include timbers for the footings, fastening, and joining parts; wedge clips and assembly instructions as specified herein.

3.2 First article. When specified (6.2.1), the contractor shall furnish a complete building for first article inspection and approval (see 4.2.1, 4.5, and 6.4).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

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3.3.1 Woods.

3.3.1.1 Lumber. Lumber of 2-inch nominal thickness and greater shall be any suitable softwood species conforming to DOC PS20; structural, dry, and dressed lumber. Lumber shall be surfaced four sides (S4S). Lumber shall have a fire-retardant treatment conforming to AWPA C20. After treatment, lumber shall be dried to an average moisture content not to exceed 15 percent.

3.3.1.2 Hardwood. Hardwood shall be straight-grained hardwood conforming to MM-L-736, type II, clear grade, flat stock, or squares.

3.3.1.3 Plywood. Plywood shall be exterior type A-C, Group I, conforming to ANSI A199.01. Plywood shall have interior type, fire-retardant treatment conforming to AWPA C27. After treatment, plywood shall be dried to an average moisture content not to exceed 15 percent.

3.3.1.4 Hardboard. Hardboard shall conform to ANSI A135.4, tempered with one side smooth (S1S).

3.3.1.5 Molding. Wood used for molding shall be kiln dried and shall conform to the Standard Molding requirements of WWPA Standard Grading Rules for western lumber.

3.3.2 Steel.

3.3.2.1 Structural steel. All structural steel shall conform to ASTM A36.

3.3.2.2 Sheet and strip. Sheet and strip steel shall conform to ASTM A242.

3.3.2.3 Galvanized steel. Galvanized steel shall conform to ASTM A526. Zinc coating after fabrication shall be in accordance with ASTM A386.

3.3.3 Guy wire. Guy wire shall conform to ASTM A475, common grade.

3.3.4 Aluminum.

3.3.4.1 Aluminum spline. The aluminum spline used in the type I building, shown on NAVFAC Drawing 1110491, shall be shaped from aluminum bar stock conforming to ASTM B308, alloy 6061, temper T4 or T6.

3.3.4.2 Aluminum brace. The 2- by 2- by 0.25-inch T-brace used in the type I building, shown on NAVFAC Drawing 1110497, shall be fabricated from aluminum stock conforming to ASTM B209, alloy 1100, temper 0.

3.3.5 Adhesives.

3.3.5.1 Gasket adhesive. Adhesive for fastening the rubber gaskets shall conform to MMM-A-187, type II. The adhesive shall be used in accordance with the manufacturer's instructions (see 3.5.7).

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3.3.5.2 Wood glue. Glue for wood assemblies shall be a phenol or resorcinol adhesive conforming to ASTM D2559 (see 3.5.6).

3.4 Parts.

3.4.1 Doors.

3.4.1.1 Personnel doors. The personnel doors shall be installed in the door panels with all hinges attached as shown on NAVFAC Drawings 1110493 and 1110494. Personnel doors shall be attached with a stainless steel, piano type, continuous hinge conforming to MS35829, Part No. MS35829-10C. The personnel doors on both the type I and type II buildings shall be identical in door size, door frame, and latching hardware as shown on NAVFAC Drawings 1110493 and 1110494. The type II building doors and door panels, shown on NAVFAC Drawings 1110520 and 1110522, shall be modified to accept the type I building door and hardware.

3.4.1.2 Overhead doors. All fabrication for installation of the overhead door operating hardware, shown on NAVFAC Drawing 1110525, shall be completed prior to delivery. Overhead doors and tracks will be installed in the field.

3.4.2 Windows. All windows shall be fitted with clear transparent, colorless, polycarbonate acrylic sheets conforming to SAE AMS3611. Windows shall be hermetically sealed with methylene chloride solvent or bonded with epoxy or urethane adhesive. The air in the cavity shall be moistureless before sealing. Sheets shall be installed with spacers as shown on NAVFAC Drawings 1110489 and 1110517.

3.4.3 Smoke pipes. The sections of the smoke pipes, shown on NAVFAC Drawings 1110502 and 1110534, shall be delivered assembled. The guy wires and smoke pipe assemblies will be installed in the roof panels in the field. When additional smoke pipes are required for the various style buildings, an equivalent number of roof panels with smoke pipe holes, shown on NAVFAC Drawings 1110500 and 1110532, shall be substituted for the plain roof panels.

3.4.4 Insect screening. Insect screening used on smoke pipe vents, shown on NAVFAC Drawings 1110502 and 1110534, shall be 18 by 14 mesh, 0.013-inch diameter aluminum wire, conforming to ANSI A201.1, natural finish.

3.4.5 Insulation. The insulation for all panels and doors shall be urethane foam conforming to ASTM C591. Urethane shall be either foamed into the panels after assembly, or may be cut with a snug fit and placed in the panels prior to fastening the exterior side plywood sheet. All insulation material used shall have a flame spread rating not greater than 25, and a smoke developed rating not greater than 50, when tested in accordance with ASTM E84.

3.4.6 Vapor barrier. A complete and tight fitting line of building paper conforming to UU-B-1780 shall be applied to the inside surface of the exterior plywood sheet of all panels, forming cups between each framing member. The vapor barrier shall not cross over the framing member.

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3.4.7 Smoke pipe insulation. The insulation around the smoke pipes, shown on NAVFAC Drawings 1110502 and 1110534, shall conform to ASTM C612, class 4.

3.4.8 Gaskets.

3.4.8.1 Type A. Type A gaskets shall be fabricated of rubber conforming to MIL-R-6130, type II, grade C, condition soft.

3.4.8.2 Type B. Type B gaskets shall be fabricated of rubber conforming to MIL-R-6130, type II, grade C, condition firm.

3.4.8.3 Type C. Type C gaskets shall be fabricated of rubber conforming to MIL-R-6130, type II, grade C, condition medium. The gasket shall have a 0.025- ±0.005-inch rubber skin conforming to MIL-R-14328, vulcanized on the curved and flat surfaces of the cellular portion.

3.4.8.4 Type D. Rubber for type D gaskets shall conform to MIL-R-900.

3.4.9 Fasteners. All fasteners shall be corrosion-resistant material or plated with corrosion-resistant material.

3.4.9.1 Nails. Nails shall conform to ASTM F547.

3.4.9.2 Screws. Wood screws shall conform to ANSI B18.6.1. Sheet metal screws shall conform to ANSI B18.6.4. Lag screws shall conform to ANSI B18.2.1.

3.4.9.3 Bolts. Bolts shall conform to ANSI B18.2.1.

3.4.9.4 Nuts. Nuts shall conform to ANSI B18.2.2.

3.4.9.5 Washers. Washers shall conform to ANSI B18.22.

3.4.9.6 Rivets. Rivets shall conform to ANSI B18.1.1 and B18.1.2.

3.4.9.7 Cotter pins. Cotter pins shall conform to SAE J487.

3.4.9.8 Chain. Chain shall conform to ASTM A466.

3.4.10 Additional fasteners. A quantity of all fasteners, in addition to the quantities specified in the bill of materials, shall be furnished with the building components as follows:

a. The additional quantity shall be 10 percent for all bolts, washers, and nuts that are under 0.5 inch in diameter.
b. The additional quantity shall be 5 percent for all bolts, washers, and nuts that are 0.5 inch in diameter or larger, and for the wedge clips.
c. The additional quantity shall be 10 percent for all screws, nails, sheet metal screws, and other required small fasteners.

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3.5 Design and construction.

3.5.1 Gasket-mounting surfaces. Surfaces to receive gaskets shall have a wood preservative coat only, and shall be entirely free of paint and any foreign matter prior to installation of gasket.

3.5.2 Tolerances. Where not otherwise specified, length and width shall be accurate within ±0.125 inch. Panels in the same group shall not differ in thickness by more than 0.063 inch, nor from design thickness by more than 0.125 inch.

3.5.3 Panel tolerances. Panels shall be square, as measured on the diagonals, within 0.125 inch for a 4-foot width, and proportionately for other widths. Panels shall be flat at all points within 0.25 inch for a 4- by 8-foot panel section and proportionately for other sizes.

3.5.4 Openings. Openings for windows, doors, and smoke pipes shall conform to specified dimensions +0.125 inch, -0.0 inch. Corners shall be square, and openings shall be square with edges of unit.

3.5.5 Wood fabrication. Surfaces of lumber shall be clean and free from oil, dust, and other foreign matter which would be detrimental to fabrication. Warp, twist, cup, or other characteristics which would prevent contact of adjacent surfaces shall not be permitted. Sanding shall not be allowed to bring framing members into tolerance.

3.5.6 Bonding or nailed-gluing. All wood assemblies shall be bonded together with an adhesive conforming to 3.3.5.2. Bonding shall be done in mechanical presses or by nailed-gluing using 6d cement-coated box nails. The nails shall be in two lines at 4-inch maximum intervals on interior framing members.

3.5.7 Gasket attachment. Gaskets shall be applied with an adhesive conforming to 3.3.5.1, and the ends shall be nailed down with 3d galvanized nails. Door gaskets shall also be stapled through the fabric backing.

3.5.8 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.5.9 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.5.10 Stenciling. The erection marks shown on the drawings shall be stenciled on each component of the building or on an attached metal or plastic

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tag, in the case of rods or other narrow items. Markings shall be gothic-type capital letters and arabic numerals not less than 1-inch high on small components, and not less than 3 inches high on panels. Stenciling shall conform to MIL-T-704.

3.5.11 Drawings. The drawings forming a part of this specification are engineering design drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the items. No deviation from the prescribed dimensions or tolerances is permissible without prior written approval of the contracting officer. In the event of a conflict between requirements on drawings and requirements specified herein, the specification shall have precedence over the drawings.

3.6 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (6.2.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.7 Erection manual. When specified in the contract, an erection manual shall be furnished (see 6.3).

3.8 Identification marking. Unless otherwise specified (6.2.1), each building shall be identified in conformance with the manufacturer's standard processes.

3.9 Standard commercial product. The assembled arctic building shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are part of the manufacturer's standard commerical product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements, manufacturer's catalogs or brochures, and represents the latest production model.

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3.10 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.11 Workmanship.

3.11.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.11.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.11.3 Assembly. The building shall be fabricated to assure ease of assembly, fit, interchangeability of like components, and proper operation of doors.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete building when a first article is required (see

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3.2 and 6.2.1). This inspection shall include the examination of 4.4 and the test of 4.5. The first article may be either a first production item or a standard production item from the contractor's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4 and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All complete buildings offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level II and the acceptable quality level (AQL) shall be 4.0 percent defective for major defects and 6.5 percent defective for minor defects. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.4 Examination. Each complete building shall be examined for compliance with the requirements specified in section 3 and table I of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign ormodification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

TABLE I. Classification of defects.

Classification	Defects	Requirement paragraph
Critical		
Major:		
101	Type/style not as specified.	1.2
102	Building not furnished as specified.	3.1
103	Materials not as specified.	3.3
104	Lumber not as specified.	3.3.1.1
105	Hardwood not as specified.	3.3.1.2

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TABLE I. Classification of defects - Continued.

Classification	Defects	Requirement paragraph
106	Plywood not as specified.	3.3.1.3
107	Hardboard not as specified.	3.3.1.4
108	Molding not as specified.	3.3.1.5
109	Structural steel not as specified.	3.3.2.1
110	Sheet and strip steel not as specified.	3.3.2.2
111	Galvanized steel not as specified.	3.3.2.3
112	Guy wire not as specified.	3.3.3
113	Aluminum spline not as specified.	3.3.4.1
114	Aluminum brace not as specified.	3.3.4.2
115	Doors not correct size, not in conformance to drawings, not installed as specified.	3.4.1.1 & 3.4.1.2
116	Windows not as specified.	3.4.2
117	Smoke pipes not in conformance to drawings; not installed as specified.	3.4.3
118	Insulation not as specified.	3.4.5
119	Vapor barrier not as specified.	3.4.6
120	Smoke pipe insulation not as specified.	3.4.7
121	Basic requirements as to design and combustion not as specified.	3.5
Minor:		
201	Gasket adhesive not as specified.	3.3.5.1
202	Wood glue not as specified.	3.3.5.2
203	Insect screening not as specified.	3.4.4
204	Gaskets not as specified.	3.4.8
205	Fasteners not as specified.	3.4.9
206	Cleaning, treatment, and painting not as specified.	3.6
207	Erection manual not furnished when specified.	3.7
208	Similar units not interchangeable.	3.10
209	Workmanship not as specified.	3.11

4.5 First article test. The complete building shall be assembled on a level surface, except that the footings shall not be included, the floor overlay shall not be cut or fastened, and the ceiling panels designed to be cut-to-fit shall not be installed. The first article shall be assembled and examined for interchangeability of like components, proper operation of all doors, and ease of building assembly without modification. The building shall then be dismantled and inspected for damage. Inability to assemble or disassemble the building without modification or damage shall be cause for rejection. After disassembly and necessary repairs have been made, the test erection unit may be submitted as a completed production unit.

4.6 Packaging inspection. The preservation, packing, and marking of the building shall be inspected to verify conformance to the requirements of section 5.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (6.2.1).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Unpainted surfaces. Unpainted and uncoated, exposed ferrous metal surfaces of component parts of the building shall be coated with type P-1 preservative. Joints of hinges shall be coated with type P-9 preservative.

5.1.1.4 Fasteners and loose items. Bolts, nuts, washers, clips, screws, nails, steel wire, and similar items needed for one complete building shall be preserved by method IC-1 or IC-3.

5.1.1.5 Truss pocket flashing, truss connectors, struts, braces, rods, roof tie assemblies, and ridge end-covers. These items of like description for one complete building shall be bundled together to form a nonshifting bundle which will provide for safe, convenient handling. The bundles shall be secured with tape conforming to PPP-T-97, type IV, or with soft annealed wire. The threaded surfaces of the rods shall be protected to prevent damage to the threads.

5.1.1.6 Smoke pipes and overhead door hardware. Smoke pipes and overhead door hardware for one complete building shall be packaged by method III.

5.1.1.7 Technical publications. The technical publications required for each building shall be packaged by method IC-3.

5.1.1.8 Consolidated packaging. Fasteners and loose items, hand tools, and technical publications for one complete building shall be consolidated andlpackaged in boxes conforming to PPP-B-636, class weather-resistant. The contents shall be cushioned, blocked, and braced to prevent movement within the boxes.

5.1.2 Commercial. The complete building shall be preserved and packaged in a manner that will insure adequate protection against deterioration and damage during shipment. Suitable commercial containers with adequate cushioning material shall be provided for components subject to damage.

5.2 Packing. The packing shall be level A or commercial, as specified (see 6.2.1).

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5.2.1 Level A. Only one complete building shall be packed as a set consisting of boxes, bundles, and crates. The bundles and crated components for each building, as specified herein, shall not exceed 5,000 pounds. Components shall be arranged within each bundle, box, and crate to provide ease of erection of the building when removed from the bundles or containers.

5.2.1.1 Lumber. Lumber shall be packed in accordance with MIL-L-14362, level A.

5.2.1.2 Plywood. Plywood shall be packed in accordance with NN-P-530, level A. Wood battens shall be applied under each girthwise strap.

5.2.1.3 Hardboard. Hardboard shall be packed in accordance with MIL-W-3448, level A.

5.2.1.4 Packaged fasteners, loose items, bundled items, smoke pipes, and consolidated packages. These items shall be packed in boxes conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or MIL-B-26195, type II, style and class optional with plywood superstructure. The contents shall be cushioned, blocked, braced, and waterproofed with a shroud in accordance with MIL-STD-1186.

5.2.1.5 Beams. Steel beams shall be bundled. The bundles shall be secured with steel strapping or bolting in combination with wood battens and blocking as required to form compact, nonshifting bundles. Strapping shall be 0.023-by 0.75-inch flat steel conforming to QQ-S-781, class 1, type I, finish B. Strapping shall be spaced not to exceed approximately 36 inches on center, with end strapping not more than 8 inches from ends. Strapping shall be stapled to any wood blocking and battens provided. The bundle shall have wooden blocks secured to the bottom of the load to provide for handling by forklift and for stacking. The height of the bottom blocks shall be not less than 2-1/2 inches.

5.2.1.6 Roof trusses. Steel roof trusses shall be bundled as specified in 5.2.1.5. Plywood covered roof trusses shall be packed in crates conforming to MIL-C-104, type I, style A, class optional. The contents shall be anchored, blocked, and braced in accordance with the crate specification.

5.2.1.7 Panels and overhead doors. All roof, wall and floor panels, and overhead doors shall be packed in crates conforming to MIL-C-104, type I, style A, class optional. Ends and edges of the panels shall be protected as necessary to prevent damage. The contents shall be anchored, blocked, and braced in accordance with the crate specification.

5.2.2 Commercial. The complete building shall be packed in a manner which will insure arrival at destination in satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The buildings are two of the basic structures intended for use in a 100-man arctic camp. Installation details for the interior furnishings and facilities are shown on NAVFAC Drawing 1290761. The buildings are designed for 100 miles per hour wind, 54 pounds per square foot snow load, and -65° Fahrenheit ambient temperature. The footings are designed for use in areas covered with snow and subject to seasonal freezing and thawing, and in permafrost regions where surface is not subject to freezing and thawing.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The type and style of building required (see 1.2).
c. When a first article is required for inspection and approval (see 3.2, 4.2.1, 4.5, and 6.4).
d. Required color of finish paint (see 3.6).
e. When identification markings are to be other than in conformance with the manufacturer's standard practice (see 3.8).
f. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph	Data Requirement	Applicable DD 1664
3.7	Manual, erection/assembly	UDI-M-24026A

(Copies of Data Item Descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

Custodian:

Navy - YD

Preparing activity:

Navy - YD

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-29165A(YD) - BUILDINGS, PREFABRICATED, PANELIZED: WOOD, ONE-ROOM, ARCTIC CLIMATE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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DATE

DD FORM 1 OCT 76 1426 Replaces edition of 1 Jan 72 which may be used. S/N 0102-LF-001-4260

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